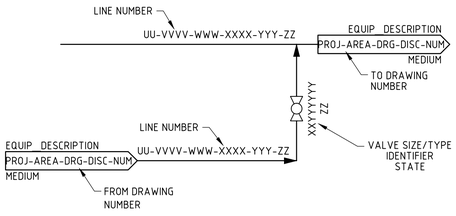
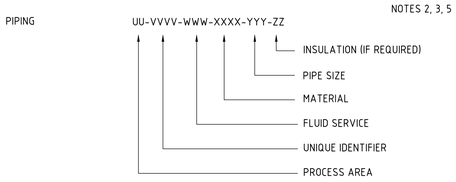
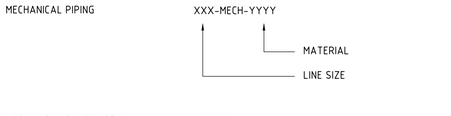
LINE DESIGNATIONS



LINE NUMBER

NOTES 2, 3, 5





STREAM NUMBERING



STREAM No (UP TO 3 DIGITS)

LINE TYPES

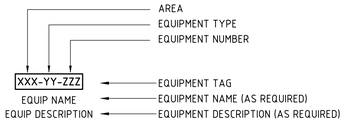
EXISTING SERVICE	
FUTURE SERVICE	
INTERMITTENT FLOW	
PRIMARY SERVICE	
SECONDARY SERVICE OR UTILITY	

INSULATION

CC	COLD CONSERVE
ET	ELECTRICALLY TRACED INSULATION
HT	HEAT TRACED INSULATION – GENERAL
PP	PERSONAL PROTECTION
ST	STEAM TRACED INSULATION
TI	THERMAL INSULATION

EQUIPMENT NUMBERING

NOTE 3



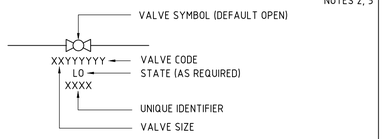
FLUID SERVICE IDENTIFICATION

NOTE 4

AA	AERATION AIR
AE	ACID EFFLUENT
AS	ANTISCALANT
BA	BLOWER AIR
BE	BARREN ELUATE
BW	BORE WATER (HYPERSALINE)
CA	COMPRESSED AIR
CN	CYANIDE SOLUTION
CO	COPPER SULPHATE
CON	CONCRETE SLURRY
COW	COOLING WATER
CR	CRUSHER OIL
CS	CYANIDE SLURRY
CT	CALCINE TAILINGS
CW	CYANIDE WATER
DCW	DECANT CYANIDE WATER
DL	DIESEL FUEL
FA	FLOTATION AIR
FD	FLOOR DRAINAGE
FL	FLOCCULANT
FR	FROTHER
FS	FLOTATION SLURRY
FT	FINAL TAILINGS
FW	FIRE WATER
GWH	GLAND WATER (HYPERSALINE)
HCL	HYDROCHLORIC ACID
HO	HYDRAULIC OIL
HP	HYDROGEN PEROXIDE
HPE	HYDROCHLORIC ACID / PREGNANT ELUATE
IA	INSTRUMENT AIR
LN	LEAD NITRATE
LO	LUBRICATING OIL
LPG	FUEL GAS
LS	LIME SLURRY
MS	MILL SLURRY
NS	CAUSTIC SODA
OG	OFF GAS
OXY	OXYGEN
PE	PREGNANT ELUATE
POT	POTABLE WATER
PSW	PROCESS SCHEME WATER
PW	PROCESS WATER (HYPERSALINE)
SC	CARBON SLURRY
SW	SHOWER WATER
TE	TERIC
TH	THERMAL OIL
TW	TRANSFER WATER
V	VENT
WM	UFG MEDIA AND PROCESS WATER
WW	WASHDOWN WATER
XA	XANTHATE

MANUAL VALVE NUMBERING

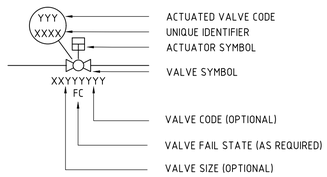
NOTES 2, 3



VALVE TYPE CODE

FLV	FLUSHING WATER WATER
GLV	GLAND WATER VALVE
RV	RELIEF VALVE
SCV	SCOUR / DRAIN VALVE
TV	TECH TAYLOR VALVE
VB	BALL VALVE
VBF	BUTTERFLY VALVE
VC	CHECK VALVE
VD	DIAPHRAGM VALVE
VDT	DART VALVE
VF	FLOAT VALVE
VG	GATE VALVE
VGL	GLOBE VALVE
VKG	KNIFEGATE VALVE
VN	NEEDLE VALVE
VP	PRESSURE / SUSTAINING VALVE
VPI	PINCH VALVE
VPL	PLUG VALVE
VR	ROTARY VALVE
VTW	3 WAY VALVE

ACTUATED VALVE NUMBERING



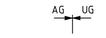
ACTUATED VALVE CODE

BDV	BLOWDOWN VALVE
DCV	DENSITY CONTROL VALVE
FCV	FLOW CONTROL VALVE
LCV	LEVEL CONTROL VALVE
PCV	PRESSURE CONTROL VALVE
PRV	PRESSURE RELIEF VALVE
PSV	PRESSURE SAFETY VALVE
SDV	SHUTDOWN VALVE (OPEN/CLOSE)
SSV	SLAM SHUT VALVE
SV	SOLENOID VALVE
TCV	TEMPERATURE CONTROL VALVE
TRV	THERMAL RELIEF VALVE
TSV	THERMAL SAFETY VALVE
XV	ISOLATION VALVE (OPEN/CLOSE)

ABBREVIATIONS

ATM	ATMOSPHERE
AG	ABOVE GROUND
AHR	AIR HOSE REEL
BDV	BLOWDOWN VALVE
BL	BATTERY LIMIT
BT	BARRED TEE
CA	CORROSION ALLOWANCE
CC	COLD CONSERVE
CCTV	CLOSED CIRCUIT TELEVISION
CONC	CONCENTRIC
DP	DIFFERENTIAL PRESSURE
ECC	ECCENTRIC
ESD	EMERGENCY SHUTDOWN
ET	ELECTRICALLY TRACED
FB	FULL BORE
FC	FAIL CLOSED
FG	FOAM GENERATOR
FH	FIRE HYDRANT
FHR	FIRE HOSE REEL
FIK	FLANGE INSULATION KIT
FL	FAIL LAST STATE
FM	FIRE MONITOR
FO	FAIL OPEN
FS	FIRE SPRINKLER
HDPE	HIGH DENSITY POLYETHYLENE
HT	HEAT TRACE
IA	INSTRUMENT AIR
LC	LOCKED CLOSED
LO	LOCKED OPEN
MIT	MONOLITHIC INSULATING JOINT
NC	NORMALLY CLOSED
NNF	NORMALLY NO FLOW
NO	NORMALLY OPEN
OD	OUTSIDE DIAMETER
O/F	OVERFLOW
O/S	OVERSIZE
PP	PERSONAL PROTECTION
PSA	PRESSURE SWING ADSORPTION
PWHT	POST WELD HEAT TREATED
RO	RESTRICTION ORIFICE
RS	REMOVABLE SPOOL
SC	SAMPLE CONNECTION
SDV	SHUTDOWN VALVE
ST	STEAM TRACED
TI	THERMAL INSULATION
UG	UNDERGROUND
U/F	UNDERFLOW
U/S	UNDERSIZE
WHR	WATER HOSE REEL

GENERAL SYMBOLS



ABOVE GROUND / UNDERGROUND



BATTERY LIMIT – SCOPE



BATTERY LIMIT – VENDOR

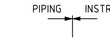
FLOW DIRECTION



HOLD DIAMOND



NUCLEAR

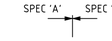


PIPING / INSTRUMENT SPECIFICATION BREAK

SLOPE – GENERAL



SLOPE – SPECIFIED

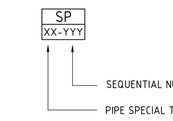


SPECIFICATION BREAK PIPE MATERIAL



TIE-IN POINT

PIPING SPECIAL ITEMS



PIPE SPECIAL TYPE

EB	EXPANSION BELLOWS
FR	FILTER / REGULATOR (AIR OR GAS SERVICES)
FS	MISCELLANEOUS FIRE SERVICES ITEMS
HS	HOSE
MC	MINSUP COUPLING OR EQUIVALENT
MS	MISCELLANEOUS ITEMS
MV	MARIC VALVE OR EQUIVALENT
OP	ORIFICE PLATE
SC	STRAUB COUPLING OR EQUIVALENT
SP	SPRAY NOZZLE
SS	SPECIAL SPOOL
ST	STRAINER

VENDOR ITEMS AND BY OTHERS



'VENDOR PACKAGE' DENOTES ALL ITEMS WITHIN BLOCK SUPPLIED BY VENDOR



'BY OTHERS' DENOTES OUTSIDE OF EPC SCOPE

NOTES

1. P&ID'S SHALL BE READ IN CONJUNCTION WITH:
   INSTRUMENT LIST
   LINE LIST
   MECHANICAL EQUIPMENT LIST
   PIPE SPECIALS LIST
   VALVE LISTS
2. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION.
3. REFER TO THE PLANT NUMBERING SPECIFICATION 2000-F00-STS-GE-10003 FOR DETAILS.
4. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION FOR PROJECT SPECIFIC FLUID SERVICE IDENTIFIERS.
5. NOMINAL PIPE DIAMETER (DN) FOR STEEL PIPES AND PIPE OUTSIDE DIAMETER (OD) FOR POLYETHYLENE PIPES.